

PIXEL-OPS Lighthouse Photogrammetry Pipeline

1. Objective

Reconstruct a 3D model of a lighthouse from overlapping UAV photographs using a custom photogrammetry pipeline built with OpenCV, NumPy, SciPy, and Open3D. No pre-built photogrammetry software (COLMAP, Meshroom) is used.

2. Input Dataset

Images: 80 high-resolution JPEG files (Lighthouse Survey DB4). EXIF focal length availability: see `dataset_report.json`. Intrinsic estimated from image dimensions when EXIF unavailable. Scale is RELATIVE (not metric) unless EXIF+sensor data confirms otherwise.

3. Pipeline Steps

- 1) Image discovery (excludes outputs/, submission/, debug/ dirs)
- 2) EXIF parsing + pinhole camera intrinsic estimation
- 3) SIFT feature extraction (up to 4000 kp/image)
- 4) FLANN matching + Lowe ratio test (threshold 0.75)
- 5) Fundamental/Essential matrix RANSAC geometric verification
- 6) Initial pair selection (max inlier count x ratio score)
- 7) Two-view init: identity + relative pose, DLT triangulation
- 8) Incremental SfM: PnP RANSAC registration + triangulation
- 9) [Optional] SciPy least_squares bundle adjustment
- 10) Semi-dense plane-sweep ZNCC depth estimation (vectorised)
- 11) Multi-view depth back-projection + adaptive fusion
- 12) Statistical + radius outlier removal (scale-adaptive)
- 13) Sparse + dense PLY export, RGB colouring, visualisation

4. Feature Extraction

Method: SIFT (Scale-Invariant Feature Transform). Total keypoints: 480,017. Mean per image: 6000. Invariant to scale, rotation, and illumination changes.

5. Feature Matching

FLANN approximate nearest-neighbour matching. Pairs matched: 863. Pairs verified: 319. Mean verified matches per pair: 249. Strategy: sequential window (configurable).

6. Geometric Verification

OpenCV `findFundamentalMat` (RANSAC, 1.5px threshold) then `findEssentialMat` + `recoverPose` for relative camera pose recovery. Minimum inlier ratio 0.20 enforced to reject degenerate configurations.

7. Structure-from-Motion

Incremental SfM with PnP RANSAC (`solvePnPRansac`). Registered: 29 cameras. Sparse points: 13,341. New 3D points triangulated for each newly registered camera. Reprojection filter applied after each registration.

8. Depth Estimation

Plane-sweep ZNCC depth estimation. References: 10. Mean valid pixels: 99.5%. Depth at 600px resolution using properly scaled camera intrinsics. OpenCV `boxFilter` for O(1) per-pixel ZNCC. Bilateral fill for gaps.

9. Point Cloud Fusion

Depth back-projection using scaled pinhole model. Raw combined: 400,000 pts. Adaptive voxel downsampling + statistical + radius outlier removal. Final dense cloud: 452,809 points (RGB coloured from source images).

10. Results

`sparse_points.ply` : 13,341 SfM sparse points
`points.ply` : 452,809 dense fused points
`depth_map_XX.npy/png` : 10 depth maps
`reconstruction_screenshot.png` : automated render
`reconstruction_report.json` : full quality metrics
`documentation.pdf` : this document

11. Limitations

- Relative scale (not metric) without GPS/GCP ground truth
- Depth is semi-dense (not full MVS); GPU would enable denser results
- Some cameras may not register if image overlap is insufficient
- Bundle adjustment disabled by default (enable: `--use-bundle-adjustment`)
- Textureless regions (sky, plain concrete) produce sparse coverage

12. Adaptability to New Dataset

Dataset-agnostic: no hardcoded filenames, image counts, or camera params. Run on any folder: `python pipeline.py --input <new_folder> --output <out> --dense`. All thresholds in `config.yaml`. EXIF auto-detected. Pair selection is automatic.